Assignment 003: Adventure

Program Report

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1 Introduction

Adventure is a C++ game. The player must find the princess and bring her back to the lobby. Meeting the monster ends the game.

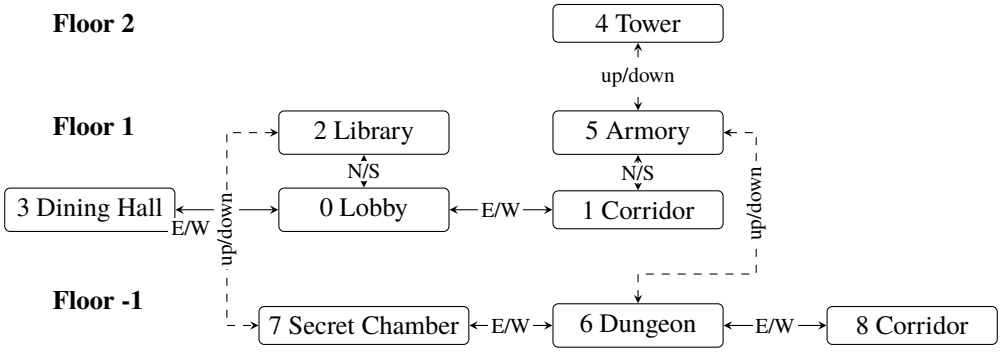
2 Program Design

Room stores the ID, type, floor, exits, and who is in the room. Its exit table links each direction to a room ID. The room type also gives its display name.

Knight stores the current room, the path taken, and whether the princess is following. Castle builds the map and places the princess and monster in different random rooms outside the lobby, while keeping a safe way to rescue her and return.

main.cpp shows the room and exits, checks commands such as go east, and moves the player. Wrong input will leave the player in the same room and the player could input again.

The map has 9 rooms of 8 types on floors -1, 1, and 2. All links work both ways.



3 Testing

The program builds with C++11. test_input.txt tests movement, and test_invalid_input.txt tests wrong input. Both tests ran as expected. Results may change because the princess and monster are placed randomly.

4 Conclusion

The game supports room movement, rescuing the princess, and checking wins and losses. Build and run instructions are in README.md.